

Clean the Water Filtration Challenge

Target grade: Grades 2-5

Time: 45 minutes - 1 hour

Team size: 3-5 students

Big Idea: Water filtration removes pollutants by passing water through layers of materials that each trap different particle sizes.

Connection: Stormwater runoff carries pollutants from roads and parking lots into salmon streams with no filtration. Students build and test a model of what filtration can do.

Learning Objectives:

By the end of this lesson students will be able to:

- Explain what stormwater runoff is and where it goes after it leaves a road or parking lot
- Describe how filtration removes particles from water using layered materials
- Design and build a water filter and evaluate its performance against other designs
- Connect the filtration challenge to real-world infrastructure like water treatment plants and rain gardens

Materials:

<u>Material</u>	<u>Quantity</u>
Plastic water bottle	1 per team
Pebbles	½ cup per team
Gravel	½ cup per team
Sand	½ cup per team
Cotton balls	6 per team
Coffee filter	2 sheets per team
Dirty water (prepared in advance) Fill a large container with water. Mix in 2-3 large scoops of potting soil, 1 scoop of sand, 2-5 drops of food coloring, and a few crushed up leaves.	Enough to fill ½ a waterbottle per team

Preparation Before the Session:

- Cut all bottles in advance
- Prepare dirty water the night before so sediment is fully mixed in
- Pre-label team stations with numbers
- Set out materials at each station before students arrive

Activity (15 minutes):

After going through the slides, give teams 15 minutes to design and build their filter. Use the final 5 minutes as a warning period and cleanup before testing begins.

Instructions to Give Students:

1. Take the top half of your bottle and flip it upside down so the bottle neck points downward. Set it into the bottom half. This is your filter on top and your collection chamber on the bottom.
2. Choose your materials and decide what order to stack them inside the top half. You can use as much or as little of each material as you want.
3. Build your filter. Pack materials firmly enough that water has to pass through them rather than around the edges.
4. When time is called, stop building and place your filter at your station ready for testing.

Guiding Questions During the Build:

- Which material do you think should go closest to the water going in at the top? Why?
- What happens if water passes through the big particles first?
- If your filter is letting water through too fast, what could you add to slow it down?
- If your filter is blocked and water is not moving through at all, what could you remove or rearrange?

TEACHER NOTE: The most effective order is gravel at the top, then sand, then cotton ball, then coffee filter at the bottle neck. Let teams discover this, and the amounts, through testing.

Test and Judge (10 minutes):

Pour identical measured amounts of dirty water into every team's filter. Give teams 5 minutes to collect filtered water, then judge on clarity of filtered water, speed of filtration, and final volume of water produced.

Discussion (10-15 minutes):

Bring the class together after judging. Ask one question at a time and let students respond freely before moving to the next.

Discussion Questions:

- Which material do you think did the most work in your filter? Why do you think that?
- What would have happened if you only used one material instead of layers?
- What other materials from real life do you think could work as a filter? Think about things you have at home or have seen outside.
- If you could redesign your filter right now knowing what you know, what would you change?
- Connect this to the real world. We just cleaned dirty water using layers of natural materials. Where do you think the water that runs off your school's parking lot goes when it rains, and is anything cleaning it before it reaches a stream?